

A Complete Bound Action–Frequency and Closed-Orbit Atlas for the Hénon Isochrone Potential

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Abstract

We close the bound action–frequency geometry of the planar Hénon isochrone Hamiltonian, with its exact circular-energy boundary. An independent apsidal integral yields the frequency map and the exact noncircular closure criterion; circular, center-crossing, escape, signed angular-momentum, and Kepler-limit faces are separated explicitly.

1 Model, conventions, and provenance

Let $\mu, b > 0$. In polar variables the unit-mass Hamiltonian is

$$H(r, p_r, L) = \frac{p_r^2}{2} + \frac{L^2}{2r^2} - \frac{\mu}{b + \sqrt{b^2 + r^2}}. \quad (1)$$

Put $\ell = |L|$,

$$A = \sqrt{\ell^2 + 4\mu b}, \quad B = \ell + A, \quad (2)$$

and use physical Hamiltonian time. For $\ell > 0$, the radial action is

$$J_r = \frac{1}{\pi} \int_{r_p}^{r_a} \sqrt{2(E - U_\ell(r))} \, dr, \quad U_\ell(r) = \frac{\ell^2}{2r^2} - \frac{\mu}{b + \sqrt{b^2 + r^2}}. \quad (3)$$

At $\ell = 0$ we use its continuous half-line limit and discuss the full Cartesian return separately.

The potential and its orbit mechanics are classical, originating in Hénon’s two 1959 papers [1, 2]. Ramond and Perez give a modern isochrone-mechanics treatment [3]; Fouvry and Prunet record the frequency map in their official Appendix C [4]. We rederive the formulas to close a reproducible certificate and make no claim of literature priority.

2 Complete bound action theorem

Theorem 1 (Energy, action, and radial period). *For fixed $\mu, b > 0$ and $\ell \geq 0$, define*

$$s_c = \frac{B^2}{4\mu}, \quad r_c^2 = s_c^2 - b^2, \quad E_c = -\frac{\mu}{2s_c} = -\frac{2\mu^2}{B^2}. \quad (4)$$

Bound motion exists exactly for $E_c \leq E < 0$. On this domain,

$$J_r = \frac{\mu}{\sqrt{-2E}} - \frac{B}{2}, \quad (5)$$

$$H(J_r, \ell) = -\frac{\mu^2}{2(J_r + B/2)^2}, \quad (6)$$

$$T_r = \frac{2\pi\mu}{(-2E)^{3/2}}, \quad \Omega_r = \frac{(-2E)^{3/2}}{\mu}. \quad (7)$$

In particular, $J_r \geq 0$ is equivalent to $E \geq E_c$ inside $E < 0$, and T_r is independent of angular momentum.

Proof. Write $s = \sqrt{b^2 + r^2} \geq b$. For $\ell > 0$, differentiation of

$$U_\ell(s) = -\frac{\mu}{b+s} + \frac{\ell^2}{2(s^2 - b^2)}$$

gives the circular equation $\mu(s-b)^2 = \ell^2 s$. Its unique solution above b is $s_c = B^2/(4\mu)$, and substitution gives (4). The effective potential tends to $+\infty$ at zero, to 0^- at infinity, and has only this critical point, hence this is its global minimum. When $\ell = 0$, the potential increases from $-\mu/(2b)$ to zero, and the same formulas give the central minimum. This proves the asserted energy domain.

For the period, set

$$x = b + \sqrt{b^2 + r^2}, \quad r^2 = x(x-2b), \quad Q(x) = 2Ex^2 + (2\mu - 4bE)x - (4\mu b + \ell^2). \quad (8)$$

A direct substitution in the energy equation gives

$$dt = \frac{x-b}{\sqrt{Q(x)}} dx. \quad (9)$$

For a noncircular bound orbit, put $q = -2E > 0$ and write $Q = q(x_a - x)(x - x_p)$. Vieta's identities give

$$x_p + x_a = 2b + \frac{2\mu}{q}, \quad x_p x_a = \frac{A^2}{q}. \quad (10)$$

The arcsine substitution on $[x_p, x_a]$ therefore yields

$$\int_{x_p}^{x_a} \frac{x-b}{\sqrt{Q(x)}} dx = \frac{\pi}{\sqrt{q}} \left(\frac{x_p + x_a}{2} - b \right) = \frac{\pi\mu}{q^{3/2}}.$$

Doubling proves (7); the circular value follows by continuity.

Finally, differentiation of the one-dimensional action integral gives $\partial_E J_r = T_r/(2\pi) = \mu/(-2E)^{3/2}$. Thus $J_r = \mu/\sqrt{-2E} + C(\ell)$. The circular cycle collapses, so $J_r(E_c, \ell) = 0$; since $\mu/\sqrt{-2E_c} = B/2$, this fixes $C = -B/2$ and proves (5). Inversion gives (6), and differentiating it in J_r gives Ω_r . $\square$

Below the circular minimum no real radial interval exists. At $E = 0$ the motion reaches the marginal escape threshold, while $E > 0$ is unbound. Equations (5) and (7) show that both action and period diverge as $E \uparrow 0$; the threshold is not a finite action point.

3 Apsidal map and noncircular closure

Theorem 2 (Frequency map and closure). *For $L \geq 0$ and $\ell > 0$,*

$$\frac{\Omega_\phi}{\Omega_r} = \beta(\ell) = \frac{1}{2} \left(1 + \frac{\ell}{\sqrt{\ell^2 + 4\mu b}} \right), \quad \frac{1}{2} < \beta < 1. \quad (11)$$

The azimuthal advance in one radial period is $2\pi\beta$. A noncircular bound phase-space orbit closes if and only if $\beta \in \mathbb{Q}$. If $\beta = p/q$ in lowest terms, its primitive period is qT_r .

Proof. Because $\dot{\phi} = \ell/r^2$, equations (9) and (8) reduce the half-cycle angular integral by

$$\frac{x-b}{x(x-2b)} = \frac{1}{2} \left(\frac{1}{x} + \frac{1}{x-2b} \right). \quad (12)$$

Besides (10), Vieta gives the shifted product

$$(x_p - 2b)(x_a - 2b) = \frac{\ell^2}{q}. \quad (13)$$

Using $\int_{x_p}^{x_a} dx/[x\sqrt{(x_a-x)(x-x_p)}] = \pi/\sqrt{x_p x_a}$ and its shifted counterpart in (12), the periapsis-to-apoapsis angle is

$$\frac{\ell}{2\sqrt{q}} \left(\frac{\pi}{\sqrt{x_p x_a}} + \frac{\pi}{\sqrt{(x_p-2b)(x_a-2b)}} \right) = \frac{\pi}{2} \left(1 + \frac{\ell}{A} \right).$$

Doubling proves the advance and (11). The same ratio follows by differentiating (6): $\Omega_\phi = \partial_\ell H$ and $\partial_\ell(B/2) = \beta$.

For a nonconstant radial phase, a phase-space return must occur after an integer number q of radial cycles. Angular return is then exactly $q\beta \in \mathbb{Z}$, equivalent to rational β ; lowest terms give primitivity. $\square$

4 Exceptional and limiting faces

Face	Exact interpretation
Circular, $E = E_c$, $\ell > 0$	The orbit is closed for every value of the epicyclic ratio. Its actual period is $2\pi/\Omega_\phi$; the noncircular rationality test does not apply.
Central equilibrium	At $\ell = 0$, $E = E_c = -\mu/(2b)$, the particle is stationary at the smooth center.
Center crossing	At $\ell = 0$, $E_c < E < 0$, the radius repeats after T_r , but velocity orientation reverses. The full Cartesian period is $2T_r$; the one-sided limiting ratio is $1/2$.
Signed momentum	Replacing $L > 0$ by $L < 0$ reverses Ω_ϕ . The geometry and radial formulas depend only on $\ell = L $.
Kepler limit	At fixed $\ell > 0$, $b \downarrow 0$ gives $J_r \rightarrow \mu/\sqrt{-2E} - \ell$ and $\beta \rightarrow 1$. The simultaneous $b, \ell \rightarrow 0$ corner reaches the Kepler collision singularity.

These clauses are logically necessary. In particular, calling every irrational-ratio circle nonclosed, or calling T_r the full Cartesian period at $L = 0$, would be false.

References

- [1] M. Hénon, “L’amas isochrone I,” *Annales d’Astrophysique* **22** (1959), 126–139; ADS bibcode 1959AnAp...22..126H.
- [2] M. Hénon, “L’amas isochrone II: Le calcul des orbites,” *Annales d’Astrophysique* **22** (1959), 491–498; ADS bibcode 1959AnAp...22..491H.
- [3] P. Ramond and J. Perez, “New Methods of Isochrone Mechanics,” *Journal of Mathematical Physics* **62** (2021), 112704; DOI [10.1063/5.0056957](https://doi.org/10.1063/5.0056957), [arXiv:2104.05643](https://arxiv.org/abs/2104.05643).
- [4] J.-B. Fouvry and S. Prunet, “Linear response theory and damped modes of stellar clusters,” *Monthly Notices of the Royal Astronomical Society* **509** (2022), 2443–2456; DOI [10.1093/mnras/stab3020](https://doi.org/10.1093/mnras/stab3020).